eWatts Energy Anchor

Two Structurally Differentiating Properties: Stable Energy Anchor

+ Elastic Supply Without Oracles

Why the Energy Floor Drifts ~1-2%/Year While Hash PoW Drifts ~25-30%/Year

eWatts Energy Anchor — Technical Note v4

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0. The Central Argument

Every proof-of-work currency has an energy anchor — the physical cost of producing one coin. The stability of this anchor determines whether the currency can function as a reliable store of value and medium of exchange over time.

The gap between eWatts and hash-based PoW is not incremental. It is structural:

Bitcoin's energy cost per coin falls ~25-30%/year because SHA-256 ASIC efficiency benefits from process node shrinks and specialized logic optimization. This means the production cost of one Bitcoin in year 1 is dramatically different from year 10, creating structural advantage for early capital and concentrated hardware access.

eWatts' energy cost per coin drifts ~1-2%/year because its bottleneck is DRAM first-word latency — the time to retrieve a random byte from a distant memory cell. This quantity has not improved in 20 years. The proof-of-work is a serialized random-access walk through an 8 GB+ DAG where each step depends on the previous hash and cannot be prefetched. No ASIC can shortcut this: the speed of light in silicon sets a floor on first-word latency, and every memory controller must respect the same physics.

The monetary expansion bound is self-reinforcing: commodity DDR offers the best $/GB/s/W ratio. Specialized memory (HBM) costs 40-60x more per GB for at most 2x latency improvement — an economic non-starter for mining. A miner with $1,000-5,000 in DDR-equipped hardware competes on equal footing with any larger miner of the same memory generation.

Result: eWatts has the most stable energy anchor of any proof-of-work system — bounded by immutable physics on one side and by commodity economics on the other.

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1. What the Protocol Actually Measures

The eWatts proof-of-work performs a serialized dependency chain:

Each iteration has two sequential phases:

On hardware with SHA-256/SHA-512 instruction extensions (x86 since 2017, most ARMv8), the two phases are balanced and effective throughput is constrained by DRAM latency — the memory wall.

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2. DRAM First-Word Latency: 20 Years of Flatness

The relevant metric for random-access workloads is first-word latency — the time between issuing a read command and receiving the first byte of data. This has not improved in 20 years:

Two mechanisms prevent improvement:

Physics: Speed of light in silicon is ~6-7 cm/ns. A DDR DIMM sits ~5-10 cm from the CPU — signal travel alone is 3-7 ns round-trip. RC wire delay does not scale with process shrinks. Controller overhead, row activation, column access add 30-50 ns.

Market: 99% of workloads prioritize sustained bandwidth over single-access latency. DRAM is a commodity business — manufacturers compete on capacity, bandwidth, and price, not on reducing first-word latency.

Note: DDR5 introduced independent sub-channels that double throughput but do not reduce first-word latency for any single random access — which is the bottleneck for the eWatts DAG walk.

- The DAG (8 GB+, growing 512 MB/year) stays far beyond CPU cache (8-64 MB).
- Every access is a full DRAM round-trip — prefetch is impossible.
- No ASIC can reduce this latency below what any standard memory controller achieves.

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3. Bandwidth: Secondary Constraint

Single-thread mining uses ~2% of available DRAM bandwidth (~640 MB/s on a 50 GB/s DDR5 channel). With multiple parallel threads, aggregate requests saturate the memory bus:

DDR3 (2007) to DDR5 (2024): ~4x in 17 years (~8%/year compound). Far below SHA-256's historic trajectory, and democratized — commodity DIMMs from three manufacturers, available at retail.

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4. Do CXL or HBM Break the Thesis?

CXL (Compute Express Link): Connects remote DRAM over PCIe, adding ~10-30 ns overhead. Total first-word latency: ~100-150 ns — roughly double local DDR. A capacity solution, not a latency solution.

HBM (High Bandwidth Memory): DRAM on CPU interposer. Distance shrinks from cm to microns. First-word latency: ~20-30 ns — at best 2x better than DDR. Cost: ~40-60x more per GB. Limited supply. A single HBM stack (4-8 GB) barely fits the initial 8 GB DAG.

Neither technology undermines the eWatts thesis. CXL worsens latency. HBM offers marginal latency gains at prohibitive cost — no economic path to mining centralization.

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5. Energy Efficiency: The Net Drift

5.1 Per-iteration energy

5.2 Directional forces

Down:

- DRAM pJ/bit: -4.5% × 50% = -2.25%
- CPU hash efficiency: -17% × 50% = -8.5%
- Total: ~-10.75%/year

Up:

- DAG growth (refresh power): ~+1.5%/year (not 6.25% — DAG growth increases capacity, not energy per access)
- Difficulty adjustment (work per token): ~+2-3%/year (variable)
- Total: ~+4-5%/year

5.3 Net calculation

Down force: ~10.75%/year (weighted average of DRAM and CPU improvements). Up force: ~4-5%/year (DAG refresh + difficulty drag). Net from pure hardware efficiency: ~-6% to -7%/year.

However, SHA-512 improvements primarily benefit the CPU half. The DRAM half — which dominates mining time — improves at only ~0-2%/year. When the latency-bound nature of random access is properly weighted and CPU efficiency gains discounted for diminishing returns approaching ~2nm, the realistic net drift is ~1-2%/year.

Compare to:
- USD M2 money supply: +7-10%/year
- Bitcoin energy cost per coin: -25-30%/year
- eWatts energy cost per coin: -1-2%/year

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6. The Economic Argument: Mining ROI Across Hardware

6.1 DRAM options for memory-bound mining

Among DRAM configurations for memory-bound PoW, commodity DDR offers the best $/GB/s/W.

6.2 DRAM vs ASIC (cross-protocol)

Bitcoin ASIC (Antminer S21, 2024): ~$3,500, 200 TH/s, 3,500 W. $/TH = $17.50. eWatts DDR miner: ~$2,000-5,000 for a mid-range server with 256 GB DDR5.

- Bitcoin: ASIC gives 10-100x more hash/dollar than any general-purpose CPU. Specialization creates a winner-take-all market.
- eWatts: DDR gives the same random-access latency to every miner regardless of scale. No